

1.4543

Alloy 455

Brand name | Custom 455®

Trade name | Alloy 455

Standards	Material No.	EN Designation	UNS	Alloy
	1.4543	X3CrNiCuTiNb12-9	S45500	455

Description | 1.4543 / Alloy 455 is a martensitic chromium stainless steel with nickel and copper addition.

Special properties | High corrosion resistance high-end hardness after aging.

Chemical Composition	C %	Si ≤ %	Mn ≤ %	P ≤ %	S ≤ %
	≤ 0.03	0.50	0.50	0.02	0.015
	Cr %	Mo %	Ni %	Ti ≤ %	Cu %
	11.0-12.5	≤ 0.50	3.00-5.00	0.90-1.40	1.50-2.50
	Nb %				
	0.10-0.50				

Mechanical Properties 20°C	Hardness HB 30 ≤ HB	0.2% Yield strength R _p ≥ N/mm ²	Tensile strength R _m N/mm ²	Elongation A ₅ ≥ %	Modulus of elasticity kN/mm ²
	480	1510	1585-1720	8	200

Physical Properties 20°C	Density g/cm ³	Specific heat capacity J/kg K	Thermal conductivity W/m K	Electrical resistivity Ω mm ² /m
	7.78	450	14	0.76

Application | Medical technology

Available forms for 1.4543 / Alloy 455	Sheets/Plates	Bars	Forged / cast parts	Finished part (drawing)
				